

Signs of Coming of Next-Gen FPGAs

Field-programmable gate arrays (FPGAs), one of the main vehicles of reconfigurable computing, is a fast developing area, with manufacturers churning out next-gen chips with greater performance and improved programmability. Here are some quick notes on new research that will mould the next-gen FPGAs

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A field-programmable gate array (FPGA) is all about programmability of software, hardware, input-output (I/O) and often analogue components too. With better design and programming tools, improved performance and energy efficiency, FPGA usage is proliferating.

Though communications remains the biggest market, FPGAs are also increasingly being used for controlling displays in consumer devices, digital signal processing, software-defined radios, radar systems, situation-aware devices, IoT products and even super-

computing. As a result, companies and universities are pooling their brains and brawns to come up with improved FPGAs, related IP, software and tools. Here is a sample of what is brewing.

Fretting over FinFET. From design companies to fabs, everybody is sweating over the FinFET transistor structure, which is believed to be the biggest development in semiconductor technology in the past four decades! FinFETs are basically 3D structures that rise above the planar substrate. This gives more volume without increasing the planar area. Plus, FinFETs are also estimated to be up to 37 per cent faster, while using less than half the dynamic power, and cut static leakage current by as much as 90 per cent, according to a technology report by Synopsys. They also help overcome performance versus power tradeoffs. Researchers have been working to reduce current leakage through the body when the device is in the state, in order to allow lower threshold voltages.

Some research groups have also shown that FinFETs can be scaled by scaling the thickness of the channel. The Korea Advanced Institute of Science and Technology, for example, has demonstrated a 3nm FinFET. It is clear that FinFET is the way ahead for

FPGAs too. Xilinx, for example has demonstrated their UltraScale range of all-programmable chips, which takes FPGA technology from 20nm planar monolithic to 16nm FinFET 3D ICs.

Fuelling FPGA innovation. Accelerated processing units (APUs) are often behind the success of FPGA designs. To this effect, AMD's upcoming Kaveri might be a notable development in the FPGA space. Kaveri pools four CPU cores and eight GPU cores and enables them to work so seamlessly that it has the power of 12 compute cores. This is their first HSA architecture based APU. They claim that adopting HSA, the new processor design standard, will enable engineers to make better use of the APUs for general computing.

Work on on-chip interfaces. It is expected that synthesisable implementations of DDR3, RDRAM, Interlaken and other high-speed interfaces that operate intra-chip and inter-chip will be important for the future of FPGAs. The Interlaken interface developed by Cisco and other original equipment manufacturers (OEMs) is popular as a server and switch interconnect for high-speed Ethernet. However, recently Cavium used Altera's Interlaken IP core to link Altera FPGAs with its own

Octeon processors, demonstrating a new way in which FPGAs with high-speed transceivers can use multi-gigabit interfaces such as Interlaken.

EDA industry leaders are all focussing on developing optimised synthesisable interfaces for systems-on-chips (SoCs). At the recently held MemCon too,

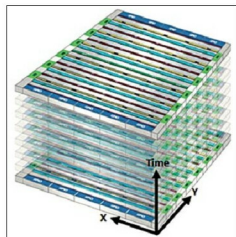


Fig. 1: Tabula's 3D architecture uses time as the third dimension (Courtesy: Tabula)

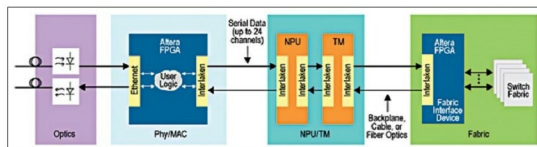


Fig. 2: Typical application block diagram using Altera Interlaken IP core (Courtesy: Altera)